Artificial Intelligence and Machine Learning

Project Report

Semester-IV (Batch-2022)

Title: - Loan Eligibility Prediction

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Abstract :

As there is rapid growth in the banking and financial sector every individual is relying on the banks and the loans provided by global and national banking sectors. In general, bank will be providing loans based on the net property value and asserts of the bank to the limited people who are eligible, and approvals of the loans will be based on the income resources of the customers.

In this project various machine learning techniques are implemented to predict the eligibility of the customer for a loan.

Customer data is collected based on various banks and accessing the customer profiles to analyse the data based on the parameters which are essential to integrate with the machine learning techniques. Analyse the data and provide the results based on the customer profile to grant the loans using machine learning approach is more advanced than traditional loan approval-based systems. The key objectives of the project are data cleansing, key attributes selection and performance comparison of SVM, Random Forest, and Logistic Regression machine learning methods on predicting the customer loan eligibility. Data is divided into training and testing parts and model is trained using train dataset and the model performance is evaluated on test dataset. Results prove that Logistic Regression algorithm has given higher accuracy as compared to other implemented models.

1. Introduction

1.1 Background

Access to credit plays a pivotal role in fostering economic growth and improving living standards. However, traditional lending practices often rely heavily on manual evaluation processes, which can be time-consuming, subjective, and prone to human bias. Inefficient credit assessment methods not only hinder financial inclusion but also pose challenges for both lenders and borrowers.

With the advent of machine learning and predictive analytics, there has been a paradigm shift in the way lending institutions evaluate loan applications. By leveraging historical data and advanced algorithms, it is now possible to automate and streamline the loan approval process, leading to faster decisions, reduced operational costs, and improved accuracy.

1.2 Objectives

1. **Develop a Predictive Model**: Design and implement machine learning algorithms to develop a predictive model for loan eligibility assessment. Explore various supervised learning techniques such as logistic regression, random forest and support vector machines to identify the most suitable model for the task.

2. **Data Collection and Preprocessing**: Gather and preprocess a comprehensive dataset containing relevant features such as applicant demographics, financial information, credit history, and loan characteristics. Perform data cleaning, imputation of missing values, and feature engineering to ensure the quality and consistency of the dataset.

3. **Feature Selection and Dimensionality Reduction**: Explore techniques for feature selection and dimensionality reduction to identify the most informative variables and reduce computational complexity. Evaluate methods such as correlation analysis, recursive feature elimination, principal component analysis, and feature importance ranking to streamline the model input space.

4. **Model Evaluation and Performance Metrics**: Assess the performance of the predictive models using appropriate evaluation metrics such as accuracy, precision, recall, F1-score. Conduct cross-validation to optimise model performance and ensure robustness.

5. **Interpretability and Explainability**: Enhance the interpretability and explainability of the predictive model to facilitate stakeholder understanding and trust. Utilise techniques such as feature importance plots, partial dependence plots, values to elucidate the factors driving loan eligibility decisions.

1.3 Significance

The loan eligibility prediction project aims to address several key challenges and deliver significant benefits:

1. **Enhanced Efficiency**: By automating the loan approval process, lending institutions can significantly reduce the time and resources required to evaluate loan applications. This efficiency gain translates into faster turnaround times for borrowers, improving customer satisfaction and retention.

2. **Improved Accuracy**: Machine learning algorithms have the potential to analyse vast amounts of data and identify complex patterns that may not be apparent through manual review. By harnessing the power of predictive analytics, lenders can make more accurate credit decisions, leading to a lower risk of default and improved portfolio performance.

3. **Promotion of Financial Inclusion**: Traditional lending practices often exclude individuals with limited credit history or unconventional income sources. By leveraging alternative data sources and advanced modelling techniques, the loan eligibility prediction model can assess the creditworthiness of underserved populations, promoting financial inclusion and expanding access to credit.

4. **Mitigation of Bias**: Human bias in the loan approval process can lead to discrimination and unfair treatment of certain demographic groups. By relying on objective data-driven criteria, the predictive model helps mitigate bias and ensure fair and consistent decision-making, fostering trust and transparency in the lending process.

5. **Cost Savings**: Manual loan processing is labour-intensive and resource-heavy, contributing to high operational costs for financial institutions. By automating credit assessment through predictive modelling, lenders can achieve cost savings, improve scalability, and allocate resources more efficiently, ultimately enhancing their competitiveness in the market.

In conclusion, the loan eligibility prediction project holds immense potential to transform the lending landscape, driving efficiency, fairness, and inclusivity in the credit market. By harnessing the power of data and analytics, we can create a more accessible, transparent, and sustainable financial ecosystem for individuals and businesses alike.

2. Problem Definition and Requirements

The problem statement for the loan eligibility prediction model is to develop a machine learning algorithm that can accurately assess the creditworthiness of loan applicants based on their demographic, financial, and credit history information. The model should automate the loan approval process, improve decision-making efficiency, and mitigate the risk of default for lending institutions while promoting fairness, transparency, and inclusivity in credit assessment.

2.1 Software Requirements:

1. **Python**: The primary programming language for implementing machine learning algorithms and data preprocessing tasks.
2. **Jupyter Notebook or Spyder**: Integrated development environments (IDEs) for Python programming and interactive data analysis.
3. **Pandas**: Python library for data manipulation and analysis.
4. **NumPy**: Python library for numerical computing and array operations.
5. **scikit-learn**: Python library for machine learning algorithms, model evaluation, and preprocessing techniques.
6. **matplotlib and seaborn**: Python libraries for data visualization and plotting.
7. **Anaconda Distribution**: Comprehensive package manager and environment manager for Python data science.

2.2 Hardware Requirements:

The hardware requirements for running the loan eligibility prediction model are relatively modest and depend on the size of the dataset and complexity of the machine learning algorithms. A standard desktop or laptop computer with the following specifications should suffice:

1. **Processor**: Intel Core i5 or equivalent processor.
2. **Memory (RAM)**: 8 GB or higher for handling large datasets and model training.
3. **Storage**: Sufficient disk space for storing datasets, code files, and software installations.
4. **Operating System**: Windows, macOS, or Linux operating system.

2.3 Data Sets:

1. **Training Data Set**: The training dataset contains historical loan application data, including features such as applicant demographics (e.g., age, gender), financial information (e.g., income, assets), credit history (e.g., credit score, previous loans), and loan characteristics (e.g., loan amount, interest rate). This dataset is used to train and evaluate the machine learning models.

2. **Test Data Set**: The test dataset contains unseen loan applications for which loan eligibility predictions need to be made using the trained machine learning model. It has the same format as the training dataset but does not include the loan eligibility outcome variable.

By addressing these requirements and utilizsing appropriate data sets, software tools, and hardware resources, the loan eligibility prediction model can be developed, evaluated, and deployed to automate and improve the loan approval process for lending institutions.

3. Proposed Design / Methodology :

1. **Problem Understanding and Data Collection:**

- **Define Objectives**: Clearly define the objectives of the loan eligibility prediction model, including the target variable (loan approval or rejection).
- **Gather Data**: Collect historical loan application data from lending institutions, including applicant demographics, financial information, credit history, and loan characteristics.

2. **Data Preprocessing and Exploration:**

- **Data Cleaning**: Handle missing values, outliers, and inconsistencies in the dataset using techniques such as imputation, outlier detection, and data validation.
- **Feature Engineering**: Create new features or transform existing ones to capture relevant information and improve predictive performance.
- **Exploratory Data Analysis (EDA)**: Conduct exploratory data analysis to understand the distribution, relationships, and patterns in the dataset using visualisations and statistical summaries.

3. **Feature Selection and Dimensionality Reduction:**

- **Feature Selection**: Identify the most informative features using techniques such as correlation analysis, univariate feature selection, or feature importance ranking.
- **Dimensionality Reduction**: Reduce the dimensionality of the dataset using techniques such as principal component analysis (PCA) or feature extraction to improve computational efficiency and reduce overfitting.

4. **Model Selection and Training:**

- **Algorithm Selection**: Choose appropriate machine learning algorithms for binary classification tasks, such as logistic regression, random forest, support vector machines.
- **Model Training**: Train multiple models using the selected algorithms on the preprocessed dataset, using techniques such as cross-validation.

5. **Model Evaluation and Validation:**
    - **Performance Metrics**: Evaluate the performance of the trained models using evaluation metrics such as accuracy, precision, recall, F1-score.
    - **Cross-Validation**: Assess model generalisation using techniques such as k-fold cross-validation to estimate performance on unseen data.
6. **Model Interpretability and Explainability:**
    - **Feature Importance**: Analyse feature importance to understand the factors driving loan eligibility decisions using techniques such as feature importance plots.
    - **Model Explainability**: Enhance model explainability using techniques such as partial dependence plots to provide insights into individual predictions.
7. **Model Deployment and Integration:**
    - **Deployment Architecture**: Design a scalable and reliable architecture for deploying the loan eligibility prediction model, considering factors such as latency, throughput, and resource utilisation.
    - **Integration with Existing Systems**: Integrate the deployed model with existing loan management systems or online platforms to automate loan approval processes and enable real-time decision-making.
8. **Monitoring and Maintenance:**
    - **Performance Monitoring**: Monitor model performance and drift over time using monitoring tools and techniques to ensure continued accuracy and reliability.
    - **Model Maintenance**: Periodically retrain and update the model using new data to adapt to changing trends, regulations, or business requirements.

4. Features

The dataset contains 13 features :

1	Loan	A unique id
2	Gender	Gender of the applicant Male/female
3	Married	Marital Status of the applicant, values will be Yes/ No
4	Dependents	It tells whether the applicant has any dependents or not.
5	Education	It will tell us whether the applicant is Graduated or not.
6	Self_Employed	This defines that the applicant is self-employed i.e. Yes/ No
7	ApplicantIncome	Applicant income
8	CoapplicantIncome	Co-applicant income
9	LoanAmount	Loan amount (in thousands)
10	Loan_Amount_Term	Terms of loan (in months)
11	Credit_History	Credit history of individual's repayment of their debts
12	Property_Area	Area of property i.e. Rural/Urban/Semi-urban
13	Loan_Status	Status of Loan Approved or not i.e. Y- Yes, N-No

5. Screenshots

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns


data = pd.read_csv("LoanApprovalPrediction.csv")
```

In [2]:
```
data.head(5)
```

Out[2]:

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History
0	LP001002	Male	No	0.0	Graduate	No	5849	0.0	NaN	360.0	1.0
1	LP001003	Male	Yes	1.0	Graduate	No	4583	1508.0	128.0	360.0	1.0
2	LP001005	Male	Yes	0.0	Graduate	Yes	3000	0.0	66.0	360.0	1.0
3	LP001006	Male	Yes	0.0	Not Graduate	No	2583	2358.0	120.0	360.0	1.0
4	LP001008	Male	No	0.0	Graduate	No	6000	0.0	141.0	360.0	1.0

In [3]:
```
data.shape
```

Out[3]:
```
(598, 13)
```

In [4]:
```
obj = (data.dtypes == 'object')
print("Categorical variables:",len(list(obj[obj].index)))
```

```
Categorical variables: 7
```

In [5]:
```
# Dropping Loan_ID column
data.drop(['Loan_ID'],axis=1,inplace=True)
```

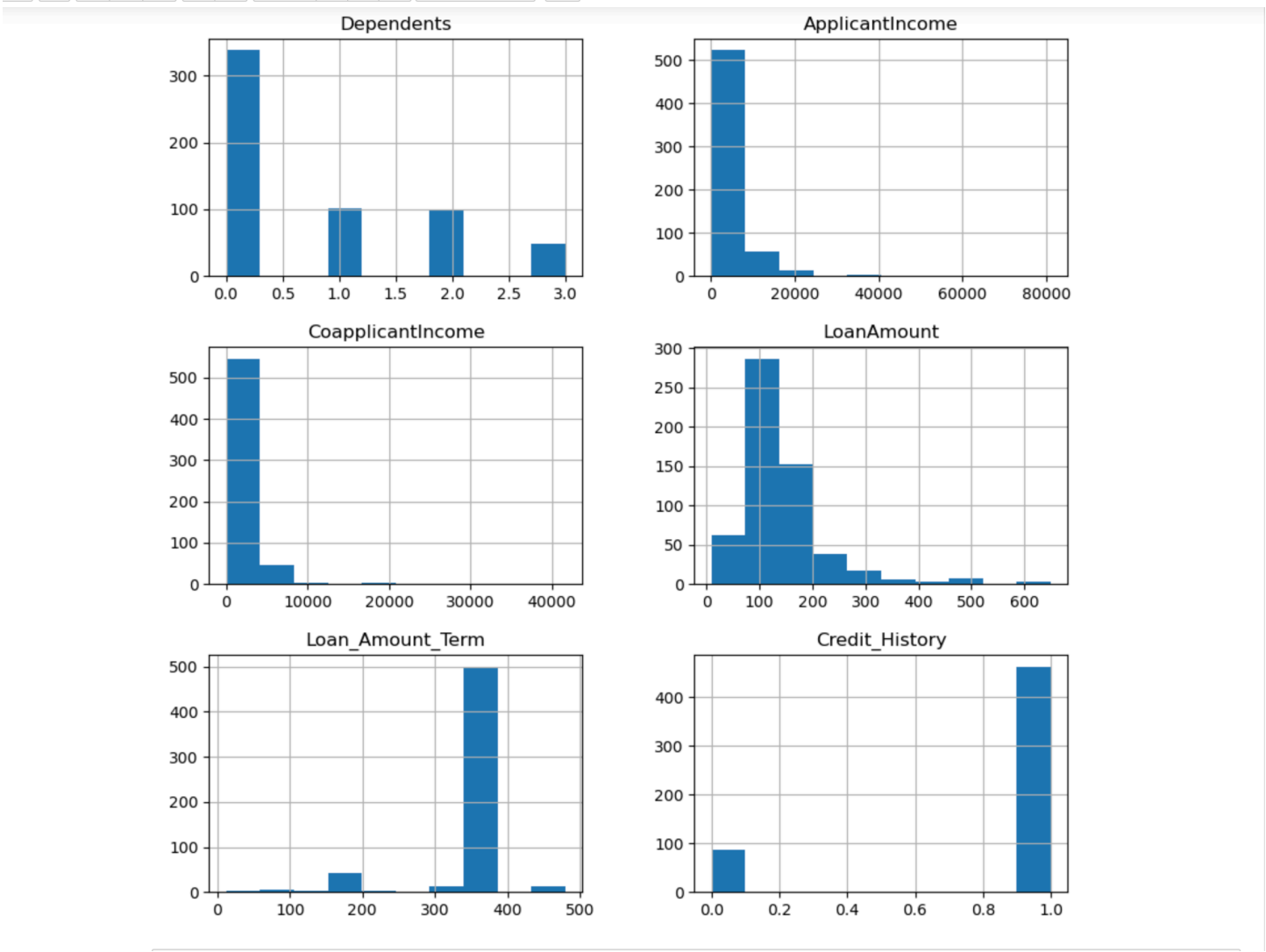
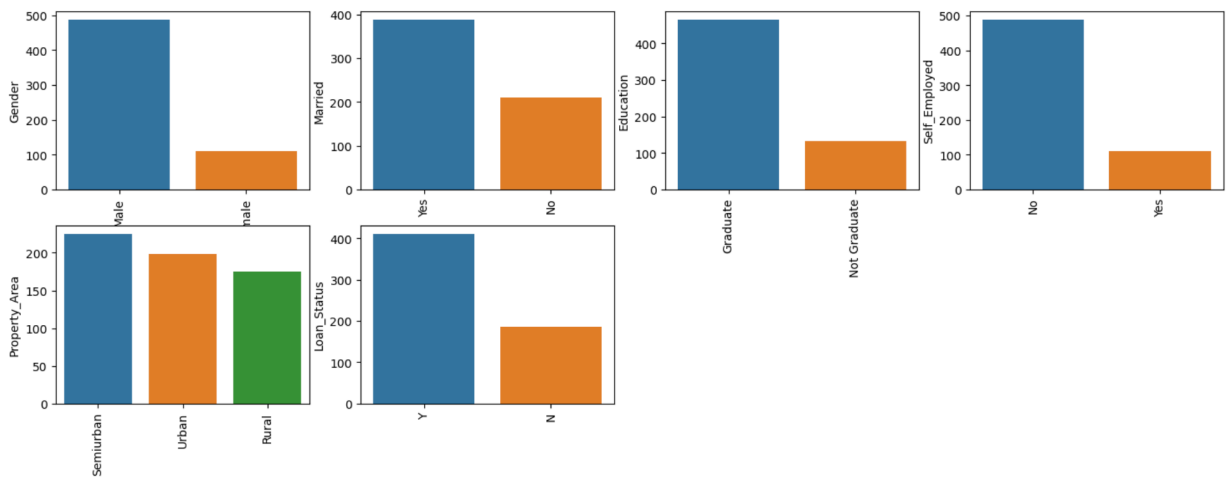


Fig. Histogram Representing the features

In [25]:
```
obj = (data.dtypes == 'object')
object_cols = list(obj[obj].index)
plt.figure(figsize=(18,36))
index = 1

for col in object_cols:
    y = data[col].value_counts()
    plt.subplot(11,4,index)
    plt.xticks(rotation=90)
    sns.barplot(x=list(y.index), y=y)
    index +=1
```

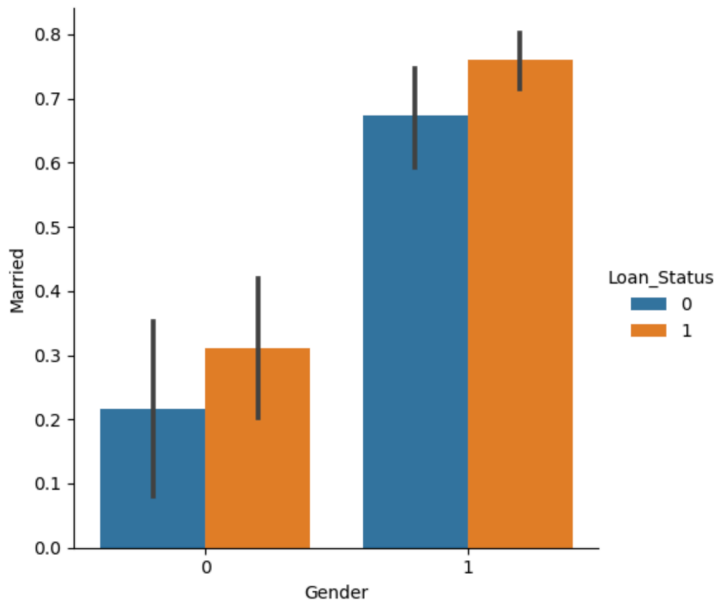


In [26]:
```
# Import label encoder
from sklearn import preprocessing

label_encoder = preprocessing.LabelEncoder()
obj = (data.dtypes == 'object')
for col in list(obj[obj].index):
    data[col] = label_encoder.fit_transform(data[col])
```

In [27]:
```
obj = (data.dtypes == 'object')
print("Categorical variables:",len(list(obj[obj].index)))
```

Fig. Barplot to visualize all the unique values.

```
In [29]: sns.catplot(x="Gender", y="Married", hue="Loan_Status", kind="bar", data=data)

Out[29]: <seaborn.axisgrid.FacetGrid at 0x7fc2acc66e80>
```




```
In [30]: for col in data.columns:
             data[col] = data[col].fillna(data[col].mean())
         data.isna().sum()

Out[30]: Gender               0
         Married              0
         Dependents           0
         Education            0
         Self_Employed        0
         ApplicantIncome      0
         CoapplicantIncome    0
         LoanAmount           0
         Loan Amount Term     0
```

Fig. Catplot to visualize plot for the Gender, and Marital Status of the applicant.

5. Correlation Matrix

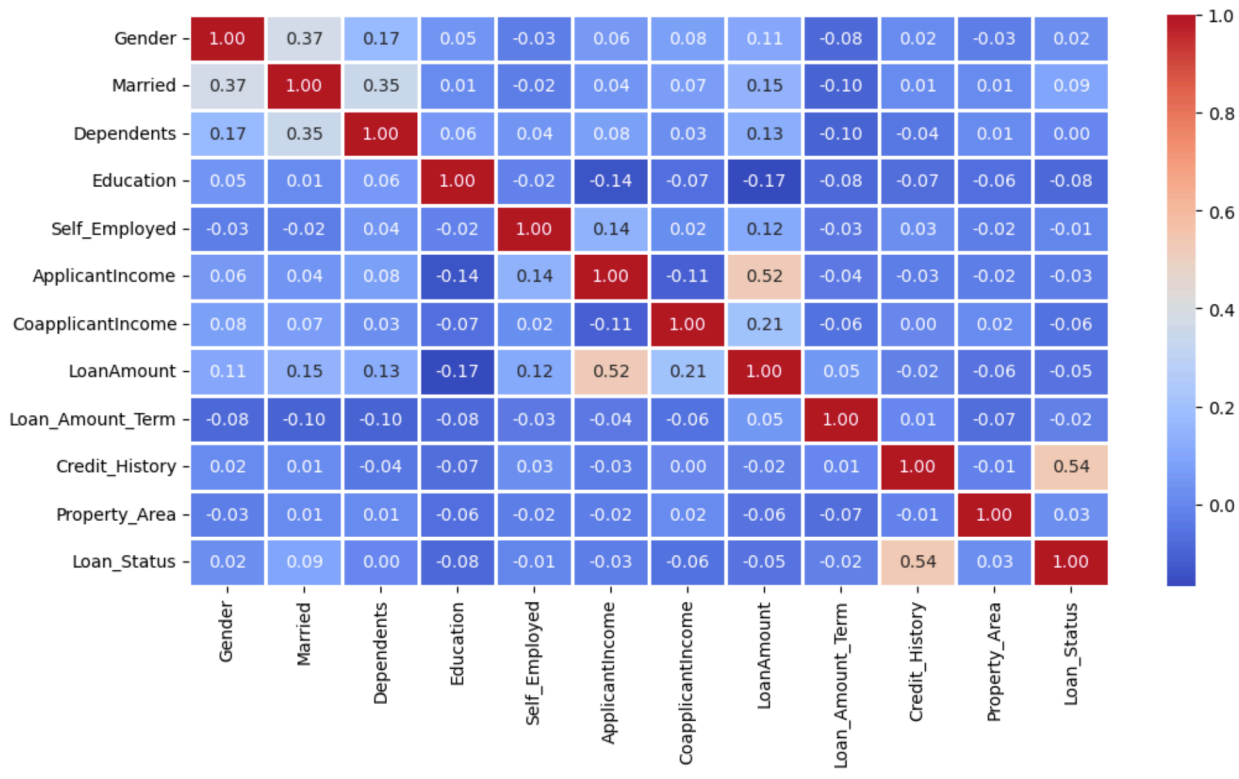


Fig. Heat Map representing the dependency between all classification

6.Result (Accuracy with different models)

In [14]:
```
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
from sklearn.linear_model import LogisticRegression
from sklearn import metrics
import warnings

# Suppressing FutureWarning from mode function
warnings.simplefilter(action='ignore', category=FutureWarning)

rfc = RandomForestClassifier(n_estimators=7, criterion='entropy', random_state=7)
svc = SVC()
lc = LogisticRegression()

# making predictions on the training set
for clf in (rfc, svc, lc):
    clf.fit(X_train, Y_train)
    Y_pred = clf.predict(X_train)
    print("Accuracy score of", clf.__class__.__name__, "=", 100*metrics.accuracy_score(Y_train, Y_pred))
```

```
Accuracy score of RandomForestClassifier = 98.04469273743017
Accuracy score of SVC = 68.71508379888269
Accuracy score of LogisticRegression = 80.44692737430168
```

In [15]:
```
# making predictions on the testing set
for clf in (rfc, svc, lc):
    clf.fit(X_train, Y_train)
    Y_pred = clf.predict(X_test)
    print("Accuracy score of", clf.__class__.__name__, "=", 100*metrics.accuracy_score(Y_test, Y_pred))
```

```
Accuracy score of RandomForestClassifier = 82.5
Accuracy score of SVC = 69.16666666666667
Accuracy score of LogisticRegression = 80.83333333333333
```

In [ ]:

Algorithms Used:

- Logistic Regression
- Random Forest
- Support Vector Machines

File Structure:

```
C:\Users\Chirag\
│
├── data/
│   ├── loanApprovalPrediction.csv
│
│
├── notebooks/
│   ├── 01_loan_eligibility_prediction.ipynb
│
```

This proposed design and methodology provide a structured approach for developing and deploying the loan eligibility prediction model, ensuring robustness, accuracy, interpretability, and scalability throughout the lifecycle of the project.

- **References :**

1. https://pandas.pydata.org/docs/
2. https://scikit-learn.org/stable/
3. https://matplotlib.org/stable/index.html
4. https://seaborn.pydata.org/